

Total No. of Printed Pages: 2

SUBJECT CODE NO: - RNP-8051
FACULTY OF SCIENCE AND TECHNOLOGY
F.E (Group A) (NEP) Sem - II
Examination May/June-2026
Basics of Electrical and Electronics Engineering

[Time: 3:00 Hours]**[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Question No. 2, 3, 4, 5, 6 and 7.

Q.1 Short Answer Questions (Solve any 10):**20**

1. Unit of current is:
 A) Volt B) Ampere C) Watt D) Ohm
2. Power is given by:
 A) V/I B) VI C) I/R D) V^2
3. Kirchhoff's voltage law states:
 A) Sum of currents zero B) Sum of voltages zero
 C) Power constant D) Resistance zero
4. Frequency unit is:
 A) Hz B) Watt C) Volt D) Ohm
5. Flux density unit is:
 A) Weber B) Tesla C) Henry D) Joule
6. Fleming's left-hand rule is used for:
 A) Generator B) Motor
 C) Transformer D) Battery
7. Transformer voltage ratio depends on:
 A) Current B) Turns ratio
 C) Resistance D) Power
8. Zener diode is used for:
 A) Amplification B) Rectification
 C) Voltage regulation D) Switching
9. Decimal system base is:
 A) 2 B) 8 C) 10 D) 16
10. OR gate output is HIGH when:
 A) All LOW B) Any input HIGH
 C) All HIGH D) None
11. Multiplexer is used for:
 A) Addition B) Selection
 C) Conversion D) Amplification

12. Strain gauge measures:

- | | |
|----------------|-------------|
| A) Temperature | B) Pressure |
| C) Strain | D) Voltage |

13. Thermistor is used for:

- | | |
|------------|----------------|
| A) Current | B) Temperature |
| C) Voltage | D) Speed |

Q.2 a) Explain resistance, resistivity and their relation. **5**

b) Describe AC and DC concepts. **5**

Q.3 a) Explain magnetic flux, mmf and reluctance. **5**

b) Describe comparison between electric and magnetic circuits. **5**

Q.4 a) Explain transformer working principle. **5**

b) Describe alternator construction and working. **5**

Q.5 a) Explain single-phase induction motor types. **5**

b) Describe universal motor. **5**

Q.6 a) Explain rectifiers and filters. **5**

b) Describe voltage regulators. **5**

Q.7 a) Explain number systems and conversions. **5**

b) Describe transducers and their applications. **5**